Human Computer Interaction (CSC-335)

Lab 06: Wireframing in Figma

Spring 2026 | Instructor: Sania Khalid | Credit Hours: 3

Lab Assessment Criteria

Criteria	Description
`Completion (30%)`	All required tasks completed within the allocated lab session time
`Quality (40%)`	Demonstrates understanding of wireframing concepts, proper use of grids, correct element hierarchy, and attention to detail
`Documentation (20%)`	Clear layer naming, organised page structure, and correctly applied styles and components
`Originality (10%)`	Original layout and creative problem-solving — not copied from others or AI-generated screens

No late submissions will be entertained. All work must be submitted before the end of the lab session.

Lab Objectives

By the end of this lab session, students will be able to:

- Set up a blank Figma project and create correctly sized frames for Desktop, Tablet, and Phone
- Apply a 12-column layout grid and row grid to a desktop wireframe frame
- Create and organise a StyleGuide page with reusable color styles
- Build standard wireframe elements: image placeholders, text blocks, section headers, and news cards
- Create a Button component with Default, Hover, and Disabled variants
- Assemble a complete desktop wireframe layout using all built elements
- Understand and apply wireframe-level fidelity: no colours, typography hierarchy through size only

---

Section 1: What is Wireframing?

A wireframe is a low-fidelity, structural blueprint of a user interface. It defines layout, spacing, and element placement without using real colours, images, or final typography. Wireframes are created early in the design process to communicate layout decisions and get feedback before investing time in high-fidelity visual design.

Wireframes serve three main purposes:

- Communication — they show stakeholders and developers what goes where without distracting with aesthetics
- Iteration speed — low-fidelity mockups can be changed in minutes compared to redesigning finished screens
- User testing — even rough wireframes can be tested with real users to validate navigation and layout decisions

Wireframe Fidelity Levels

Low-fidelity (Lo-Fi) — Boxes, placeholder text, no colour. Fastest to create. Used in early stages.

Mid-fidelity — Real text, basic spacing, greyscale. Used for layout reviews.

High-fidelity (Hi-Fi) — Full colour, real images, final fonts. Matches the finished product.

In this lab we work at Low-fidelity level only.

Section 2: Project Setup — Frames & Pages

Every Figma wireframing project starts with a clean blank file containing the right frames and a structured page layout. Follow these steps exactly.

2.1 Creating a New Blank Project

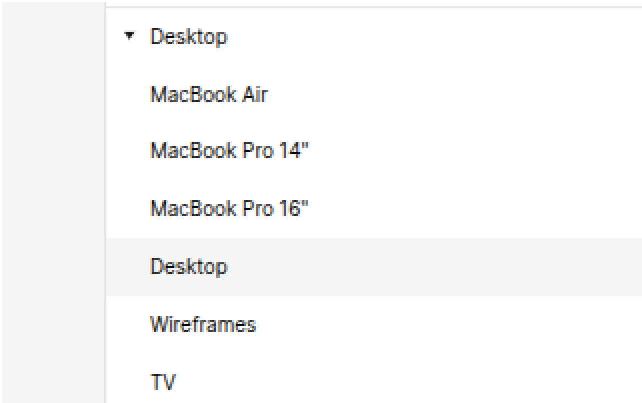
1. Open Figma and click + New design file from the home screen
2. The file opens with a blank canvas — nothing is on it yet
3. Double-click the file name at the top of the screen and rename it: HCI-Lab06-Wireframes-YourName

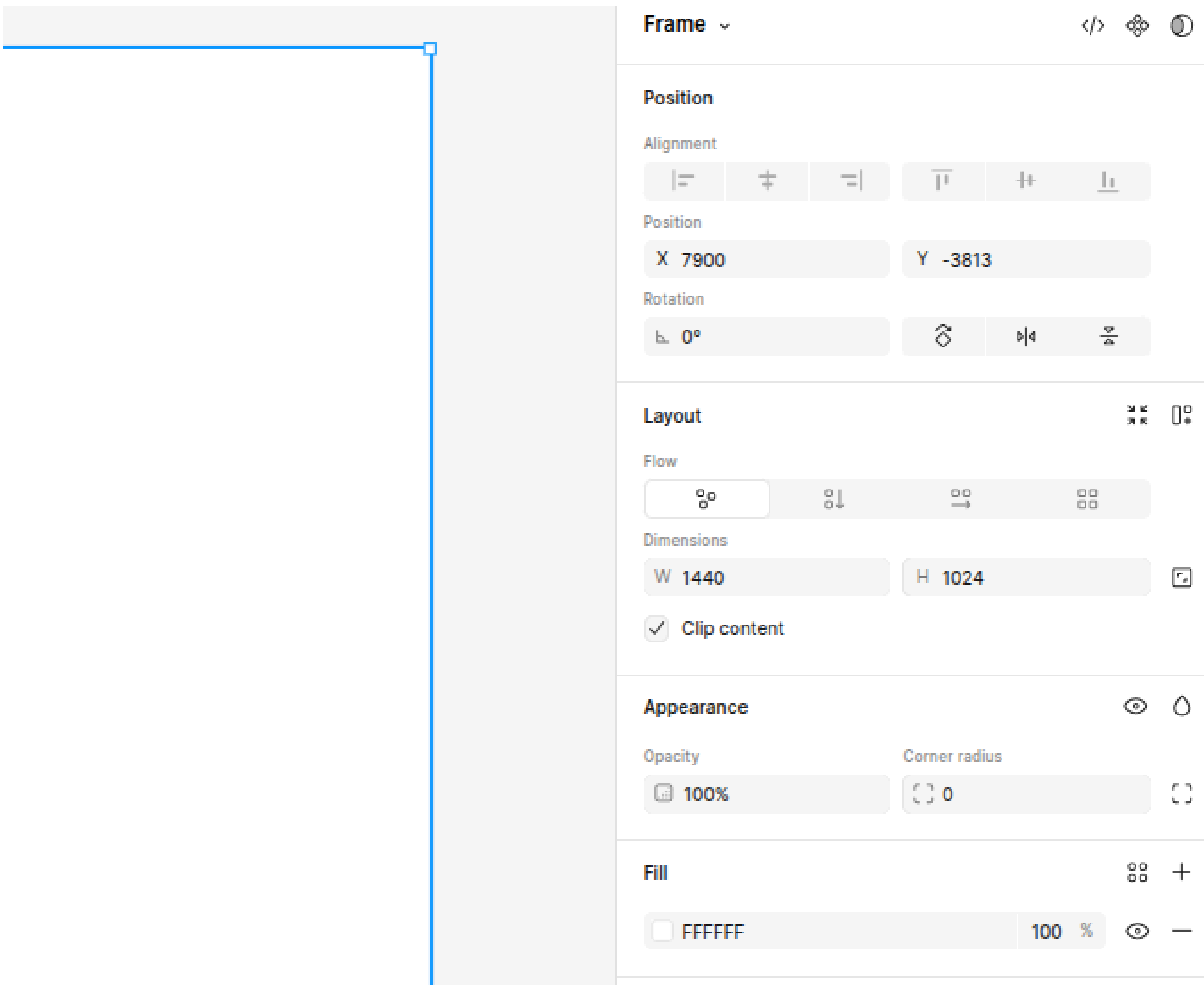
2.2 Adding Device Frames

Press F (or click the Frame tool in the toolbar) to activate the Frame tool. On the right panel you will see preset frame sizes grouped by device type.

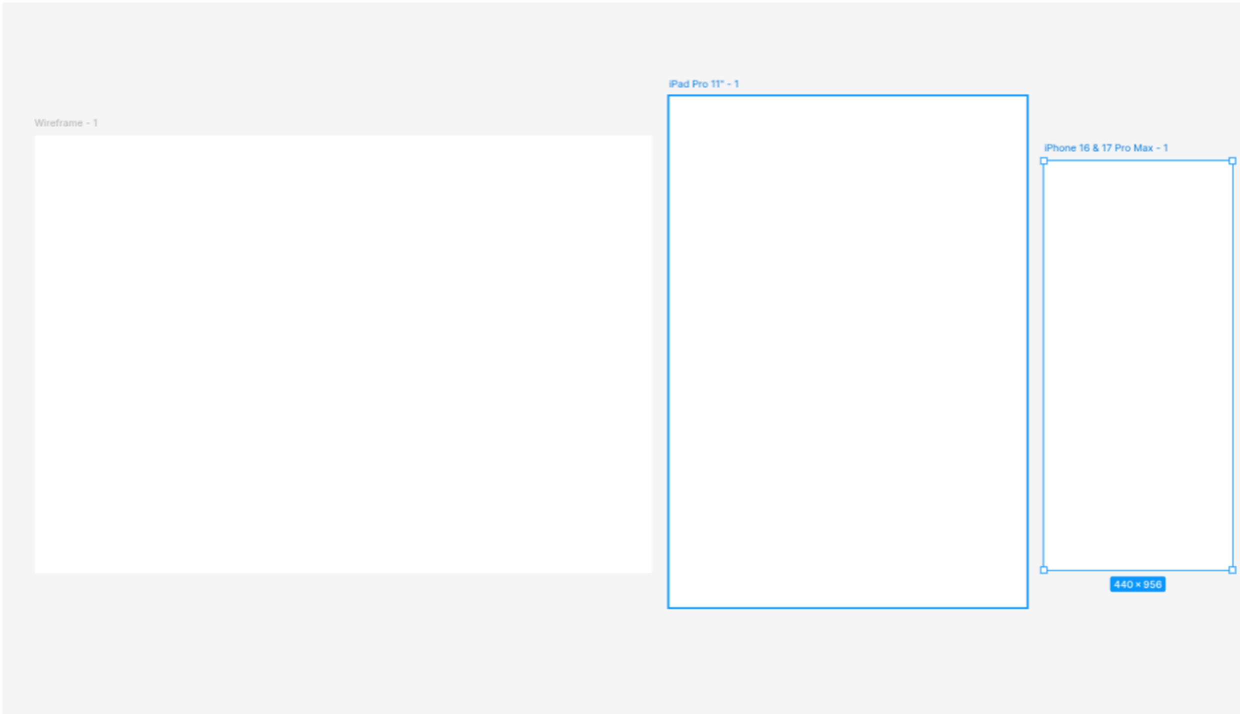
Add three frames — one for each device:

Device	How to Add
`Desktop Wireframe`	Select: Desktop > Wireframes (1440 × 1024 px) — for the main layout practice
`Tablet`	Select: iPad Pro 11" or similar (834 × 1194 px) — for the responsive tablet view
`Phone`	Select: iPhone 16 & 17 Pro Max (440 × 956 px) — for the mobile view





After adding all three frames, your canvas should look like this:

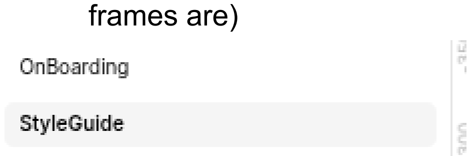


Section 3: StyleGuide Page — Colors & Styles

Before placing any elements on your wireframe frames, create a dedicated StyleGuide page. This page stores all your reusable color styles so they are available across every page in the file.

3.1 Creating the StyleGuide Page

1. In the left panel, click the + icon next to Pages to add a new page
2. Rename the page: StyleGuide (double-click the page name to rename it)
3. Rename the original page: OnBoarding (or leave it as Page 1 — this is where your wireframe frames are)



3.2 Creating Color Styles

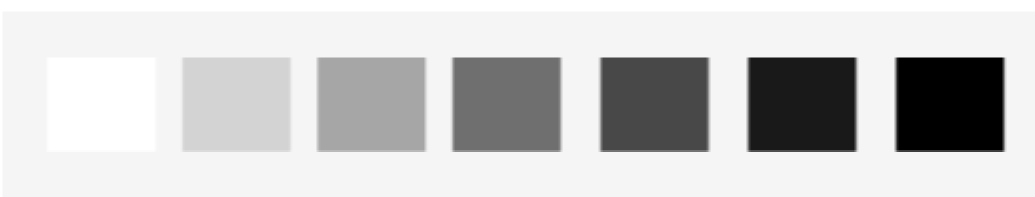
On the StyleGuide page, create a row of 7 rectangles representing your wireframe greyscale palette. Then save each one as a named Color Style.

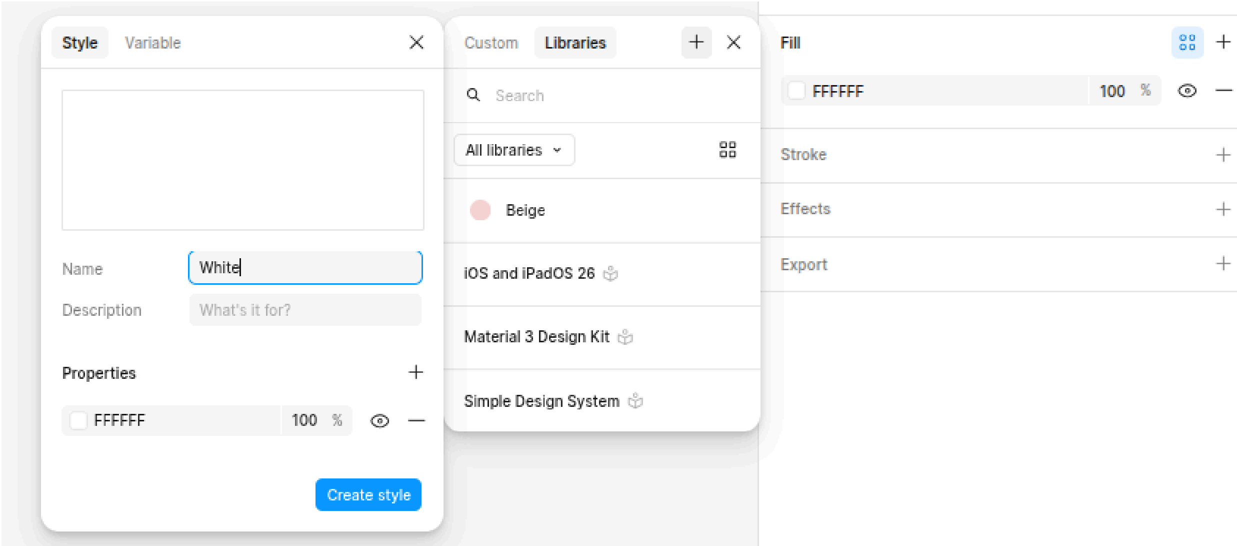
Greyscale palette to create (7 colours, left to right):

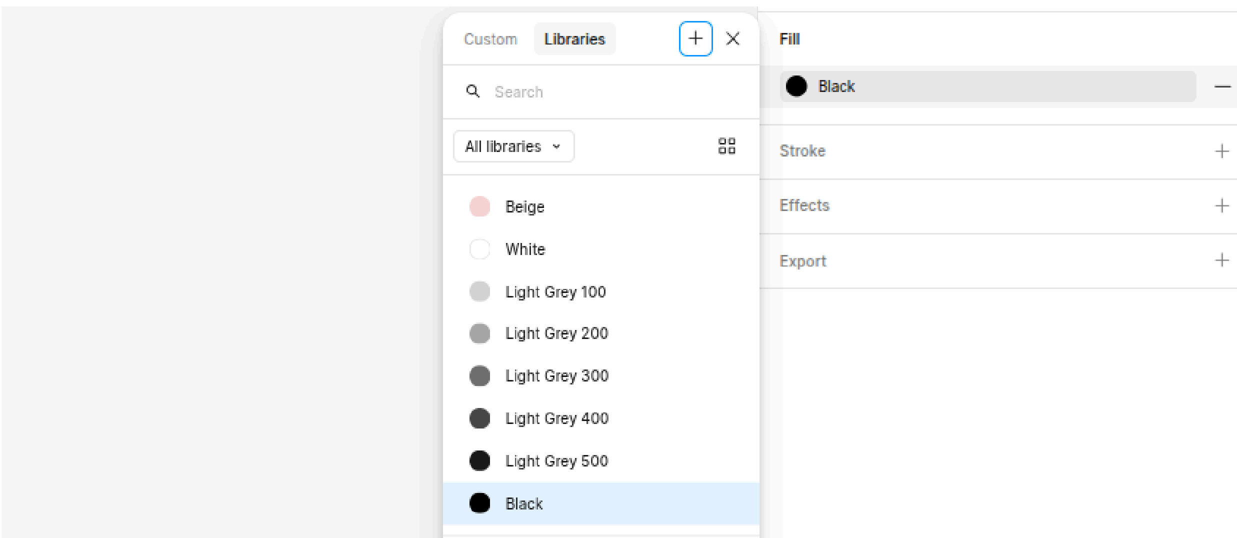
Style Name	Hex Value & Usage
`White`	#FFFFFF — pure white, used for backgrounds and card surfaces
`Light Grey 100`	#E8E8E8 — very subtle tint, used for alternate row backgrounds
`Light Grey 200`	#D0D0D0 — light grey, used for placeholder text bars
`Light Grey 300`	#AAAAAA — medium light, used for image placeholder fills
`Light Grey 400`	#787878 — medium grey, used for secondary placeholder elements
`Light Grey 500`	#444444 — dark grey, used for headline and button fills
`Black`	#000000 — pure black, used for body text in text blocks

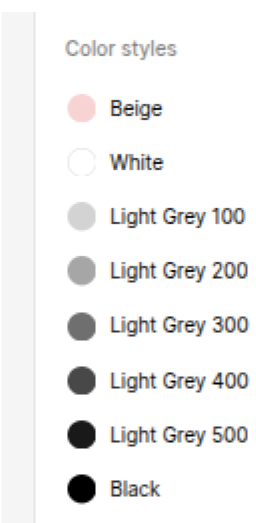
How to Create a Color Style

1. Draw a rectangle on the StyleGuide canvas (e.g., 80 × 80 px) and set its fill to the target colour
2. With the rectangle selected, click the four-dot grid icon next to Fill in the right panel
3. In the dialog, click + (Create style) — name it exactly as shown in the table above
4. Repeat for all 7 colours




Style
Variable
Name
White
Description
What's it for?
Properties
FFFFFF
100 %
Create style
Custom
Libraries
Search
All libraries
Beige
iOS and iPadOS 26
Material 3 Design Kit
Simple Design System
Fill
FFFFFF
100 %
Stroke
Effects
Export


Custom
Libraries
Search
All libraries
Beige
White
Light Grey 100
Light Grey 200
Light Grey 300
Light Grey 400
Light Grey 500
Black
Fill
Black
Stroke
Effects
Export


Color styles
Beige
White
Light Grey 100
Light Grey 200
Light Grey 300
Light Grey 400
Light Grey 500
Black

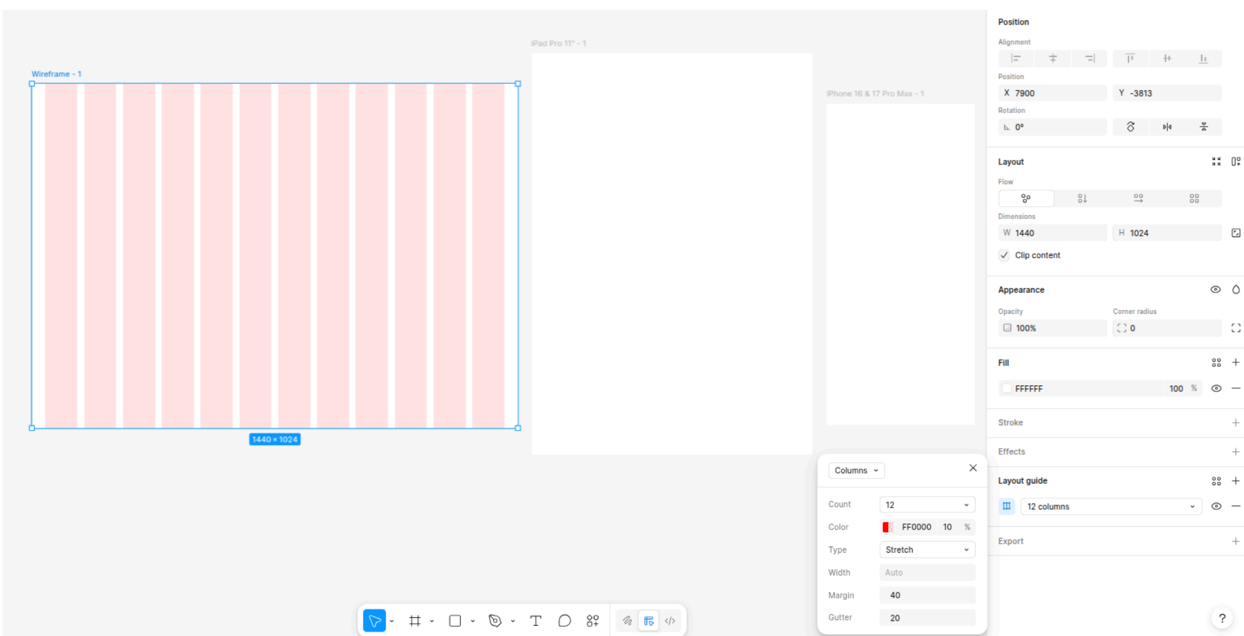
Section 4: Layout Grid

A layout grid helps you align elements consistently across your wireframe. We apply two grids to the desktop frame: a 12-column grid for horizontal alignment, and a row grid for vertical rhythm.

4.1 Applying a 12-Column Grid

1. Select the Desktop Wireframe frame on the canvas (click its border, not a child element)
2. In the right panel, scroll to Layout guide and click + to add a grid
3. Click the grid type icon (shows a square grid by default) and switch it to Columns
4. Set the following values:

Setting	Value
`Count`	12
`Type`	Stretch
`Margin`	40 px (left and right gutters from the frame edge)
`Gutter`	20 px (space between each column)
`Color`	Set to a soft red/pink — e.g., #FF000010 at 10% opacity — to see the grid clearly

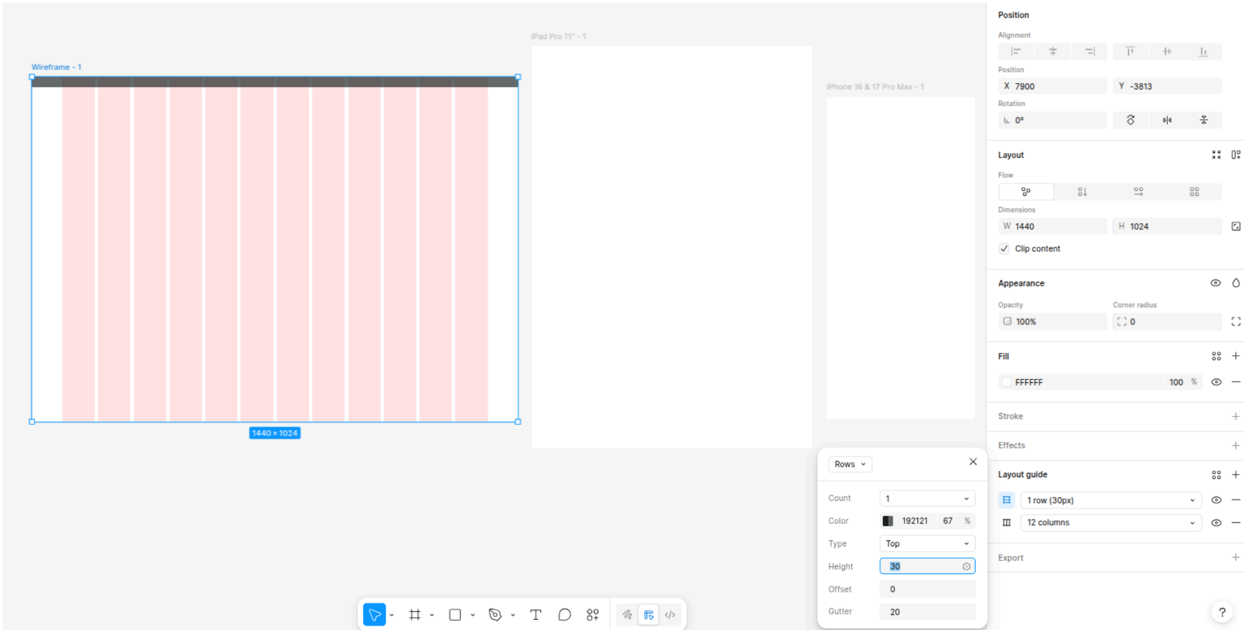


4.2 Adding a Row Grid

Add a second grid entry to the same frame for vertical alignment:

1. With the Desktop frame still selected, click + in Layout guide again
2. This time select Rows from the grid type options

3. Set: Count = 1, Type = Top, Height = 80 px (this marks the navigation bar height), Gutter = 20 px



Why Two Grids?

The column grid ensures every element snaps to a consistent horizontal rhythm (like CSS flexbox columns).

The row grid marks fixed regions — typically the navigation bar — so you never accidentally place content

where the nav bar will sit.

Section 5: Building Wireframe Elements

Now we build the reusable wireframe elements that will be placed into the layout. All of these will be created on the StyleGuide page first and then used (or copied) into the wireframe frames.

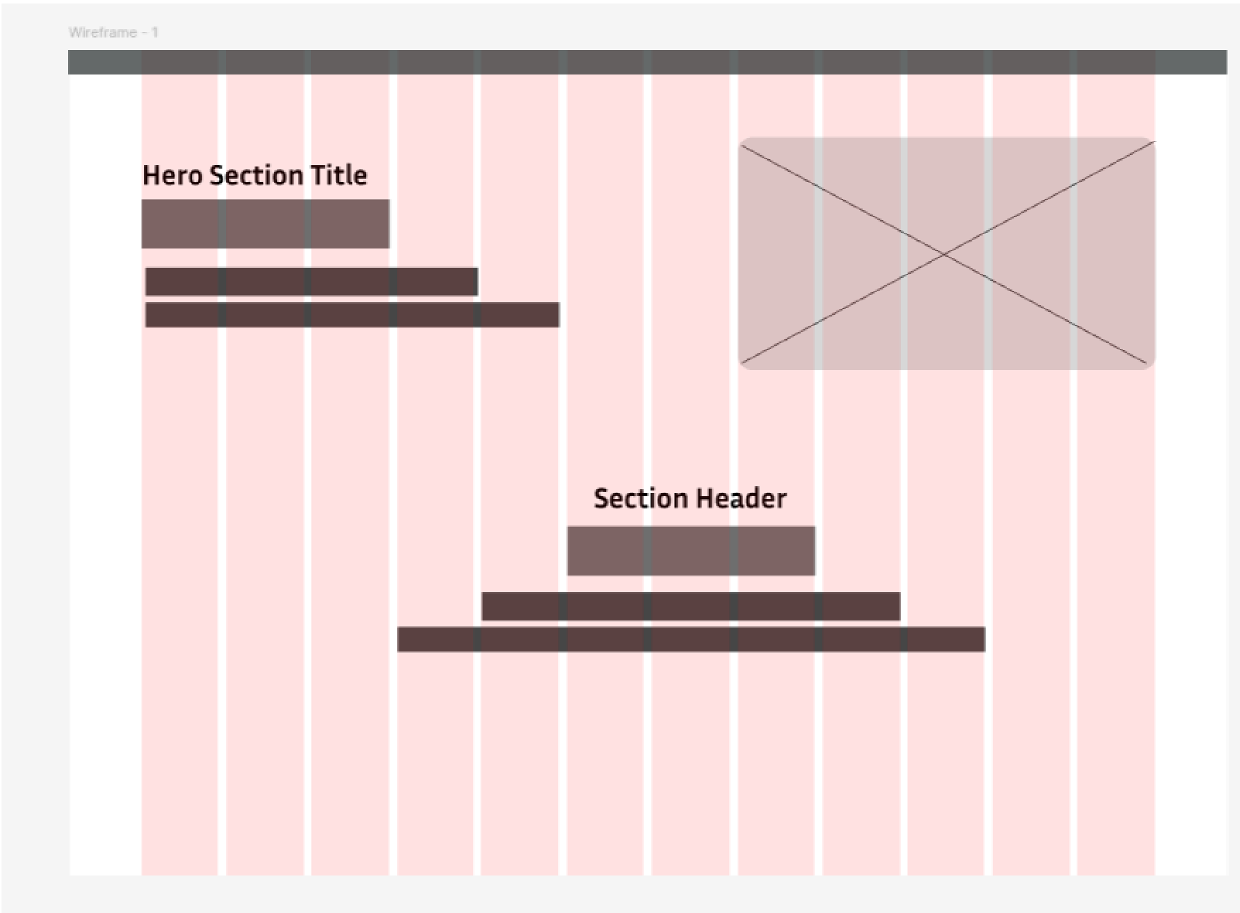
5.1 Image Placeholder — Rectangle with X Cross

The classic wireframe image placeholder is a rectangle filled with Light Grey 300, with two diagonal lines drawn corner-to-corner forming an X. It communicates 'an image goes here' without using any real image.

How to Build It

1. Draw a Rectangle (R) at your desired image size (e.g., 400 × 280 px)
2. Set fill to Light Grey 300 color style

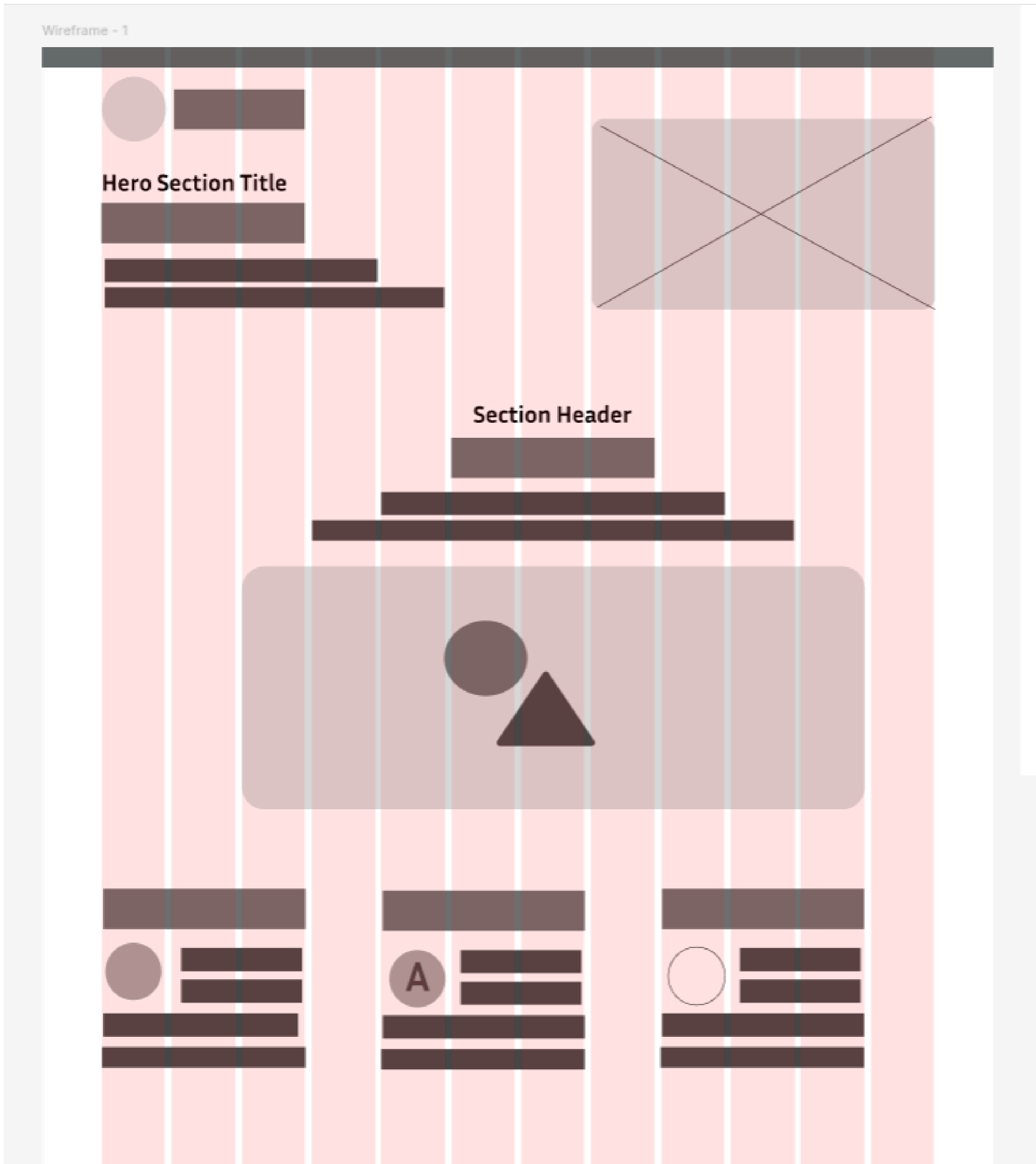
3. Draw a Line from top-left corner to bottom-right corner (use the Line tool or Pen tool)
4. Draw a second Line from top-right corner to bottom-left corner
5. Set both lines to Light Grey 400, stroke weight 1.5 px
6. Group all three shapes (Ctrl + G) and name the group: Image Placeholder
7. Convert the group to a component (Ctrl + Alt + K)



5.2 Image Placeholder with Circle + Triangle

A second image placeholder variant uses a circle and triangle to suggest a photo/media icon inside the rectangle. This is used for content image zones (like video thumbnails or featured articles).

4. Duplicate your Image Placeholder component
5. Remove the X lines from the duplicate
6. Add an Ellipse (O) in the centre-bottom area, fill Light Grey 400, e.g., 40 x 40 px
7. Add a Polygon (triangle) adjacent to the ellipse, fill Light Grey 500, e.g., 40 x 40 px
8. Group and name it: Image Placeholder (Media)



5.3 Text Blocks — Hero Title & Section Header

Text blocks in wireframes use horizontal grey bars to represent text lines. Each bar's height corresponds to the text size — taller bars for headings, shorter for body.

Logo / Avatar placeholder

4. Draw a small Circle (O), ~40 x 40 px, fill Light Grey 300

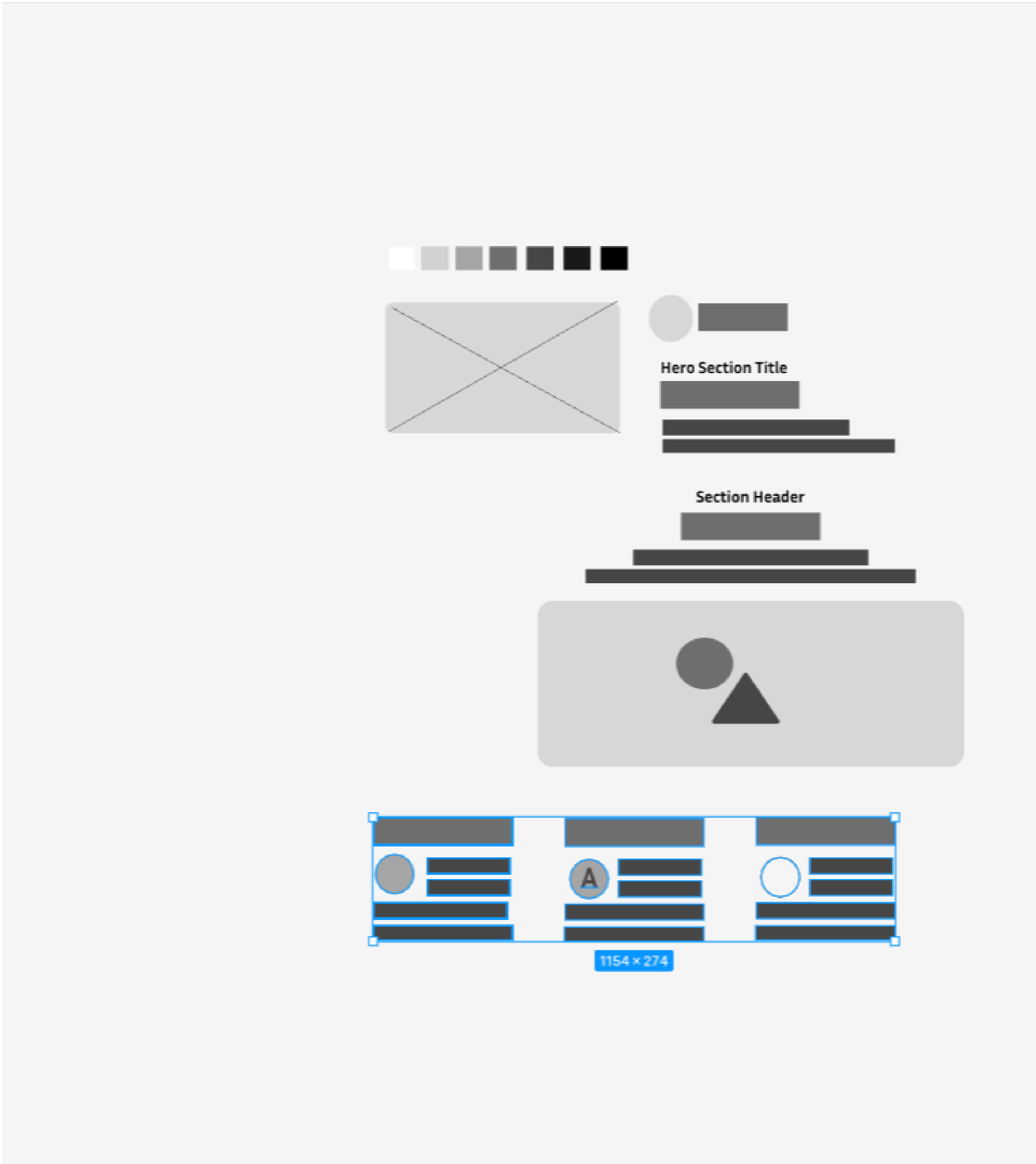
5. Place a Rectangle beside it (~120 x 20 px), fill Light Grey 400 — this represents a company name
6. Group them and name: Logo

Hero Section Title Block

5. Draw a Rectangle for the H1 heading line: width ~280 px, height 28 px, fill Light Grey 500
6. Below it, add a shorter bar: width ~200 px, height 10 px, fill Light Grey 300 — subheading line
7. Add two more body text bars: width ~320 px and ~360 px, height 8 px, fill Light Grey 300
8. Stack with 8 px gap between bars using Vertical Auto Layout (Shift + A)
9. Name the frame: Header H1 Placeholder

Section Header Block

5. Create a centred stack: heading bar (240 px wide, 22 px tall) + two supporting text bars
6. Set alignment to Center in the Auto Layout settings
7. Name it: Section Header Placeholder



5.4 News / Content Card

Content cards appear in grids and feeds. Each card contains an avatar (circle), a title bar, and 3-4 body text bars arranged vertically.

4. Draw a Circle (~40 x 40 px), fill Light Grey 300 — represents the avatar/profile picture
5. To the right, add a title bar: width ~180 px, height 14 px, fill Light Grey 500
6. Below the title, add 3 text bars: widths 200, 160, 140 px, height 8 px, fill Light Grey 300
7. Arrange avatar and text stack side by side using Horizontal Auto Layout, gap 12 px
8. Wrap the whole card in a frame with 16 px padding and name it: Card With Icon

9. Convert to a component (Ctrl + Alt + K)

5.5 Paragraph Placeholder

The paragraph placeholder is a simple stack of 3-4 grey bars representing body copy text. Use it wherever descriptive text appears.

8. Create 3 rectangles: heights 8 px, widths 320 px, 300 px, 280 px, fill Light Grey 300
9. Stack with Vertical Auto Layout, gap 6 px
10. Name: Paragraph Placeholder

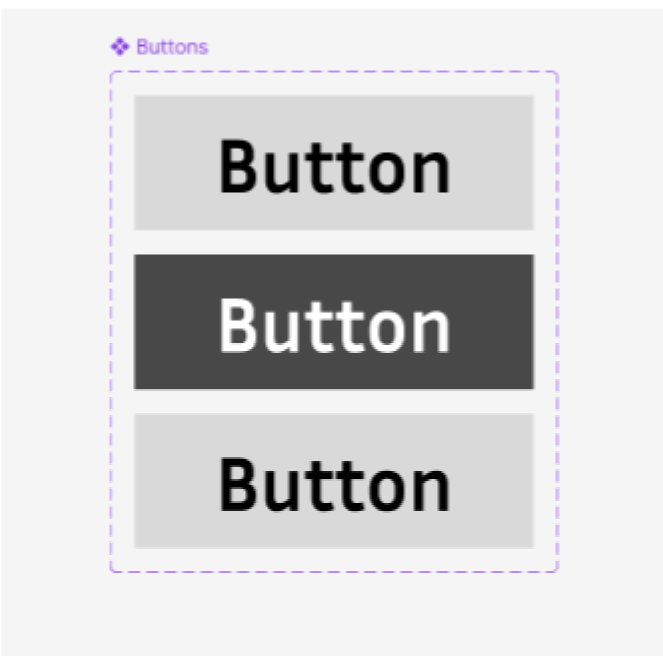
11. Convert to a component

Section 6: Button Component with Variants

Buttons in wireframes are solid filled rectangles with a text label. We create a Button component with three variants: Default, Hover, and Disabled — so instances on any screen can switch between states.

6.1 Creating the Base Button

9. Draw a Rectangle: width 180 px, height 44 px
10. Set fill to Black color style
11. Add a Text layer on top: type 'Button', colour White, font size 16, centred
12. Group the rectangle and text (Ctrl + G), then convert to component (Ctrl + Alt + K)
13. Name the component: Button

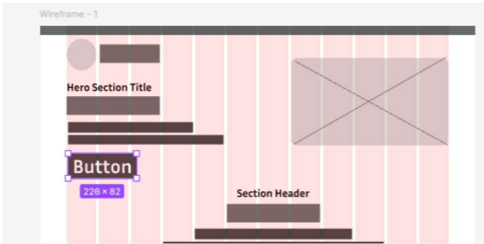


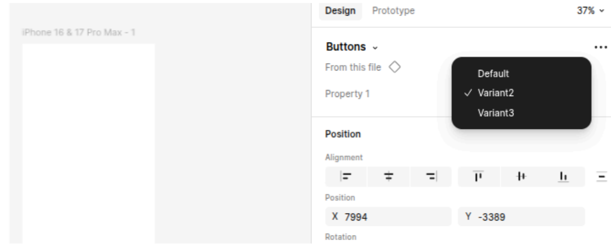
6.2 Adding Variants (Default, Hover, Disabled)

7. With the Button component selected, click 'Add variant' in the right panel (or right-click > Add variant)
8. Two variants appear inside a purple dashed box — this is the Component Set
9. Select all variants, click 'Add variant' once more — now you have 3 variants
10. Rename the property: in the right panel, change 'Property 1' to 'State'
11. Name each variant by clicking on it and editing its State value:

- State = Default — Keep original black fill, white text

- State = Hover — Change fill to Light Grey 500 (dark grey), keep white text
- State = Disabled — Change fill to Light Grey 200, change text colour to Light Grey 400





Key Concept: Component Set vs Component

A single component has one state. A Component Set groups multiple variants under one parent. When you place an instance of a component set on your canvas, you can switch its 'State' property from the right panel — without replacing the element or breaking Auto Layout.